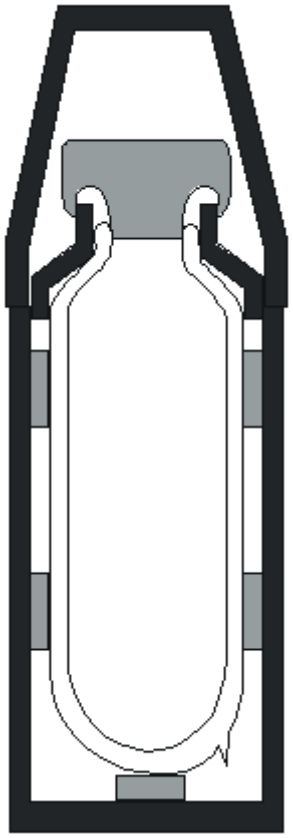
Q1.

The diagram below shows a vacuum flask.



(a) Give **two** features of the flask which reduce heat loss by conduction.

1. _____

2. _____

(2)

(b) Give **one** feature of the flask which reduces heat loss by radiation.

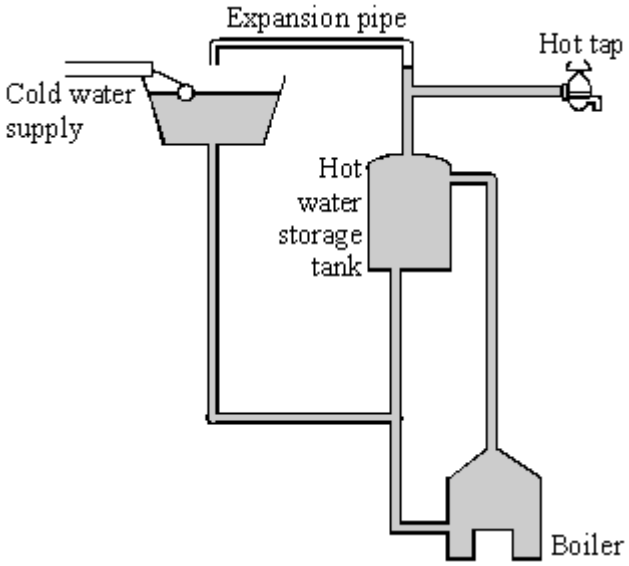
_____

(1)

(Total 3 marks)

Q2.

(a) The diagram shows a hot water system.



(i) Explain why the boiler is below the hot water tank.

_____

_____

_____

(ii) Why is heat energy transferred from hot water in the tank to the surrounding air?

_____

_____

(iii) Name the process by which energy is transferred through the sides of the tank.

_____

(iv) How may heat loss from the hot water tank be reduced?

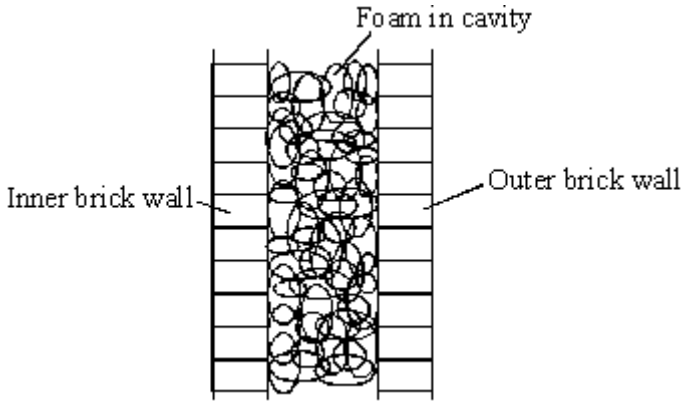
_____

_____

_____

(6)

(b) One way of reducing heat loss from a house is by cavity wall insulation. Foam is pumped between the inner and outer brick walls as shown in the diagram.



How is heat loss from a house reduced by:

(i) having a cavity wall?

_____

_____

_____

_____

(ii) filling the cavity with foam?

_____

_____

(3)

(Total 9 marks)

Mark schemes

Q1.

(a) plastic/glass walls; vacuum; insulating top

any two for 1 mark each

2

(b) silvering/shiny on either wall

for 1 mark

1

[3]

Q2.

(a) (i) hot water rises (not heat)

for 1 mark

due to convection currents
or water expands/becomes less dense on heating
or less dense water rises

any for 1 mark

2

(ii) inside hotter (than outside)

for 1 mark

1

(iii) (heat transfer by) conduction

for 1 mark

1

(iv) surround/cover/insulate tank with poor conductor **or** named insulator

for 1 mark each

2

(b) (i) air is an insulator/poor conductor

for 1 mark

1

(ii) convection stopped foam is an insulator/poor conductor

for 1 mark each

2

[9]